Smartphone-based Indoor Localization

Danish Vasdev
Nanyang Technological University
G2607761K
Singapore
danish017@e.ntu.edu.sg

Himadri Khanchandani
Nanyang Technological University
G2610348A
Singapore
himadri001@e.ntu.edu.sg

Kusalavan Kirubhaharini
Nanyang Technological University
G2503701C
Singapore
kirubhah002@e.ntu.edu.sg

Pramod Sanjay
Nanyang Technological University
G2505986D
Singapore
sanjay009@e.ntu.edu.sg

***Abstract*—Indoor localization aims to provide accurate and reliable location-based service within buildings using various sensor data. Smartphones have become useful for indoor localization due to their widespread usage and the variety of sensors embedded in them. In this report, raw trace file data containing ground-truth positions, geomagnetic signals, WiFi signal strength, and iBeacon data is cleaned, preprocessed, analyzed, and visualized. Various techniques are explored to predict indoor positions, including inertial sensor-based position estimation, deep learning-based location fingerprinting, and sequence-based localization. Each approach uses different sensor information and estimation strategies to predict device positions within indoor environments.**

***Index Terms*—urban computing, indoor localization**

I. Introduction

Conventional GPS technology helps to determine the location of a device or object on Earth. Using a technique called tri-lateration, the device collects signals from multiple satellites to determine the exact coordinates. However, the accuracy of GPS is drastically affected by building materials like concrete, brick, and stone, which reflect or absorb high-frequency radio waves. These limitations of GPS have motivated the development of indoor localization techniques for tracking and navigating devices/objects inside building structures.

Multiple sensor data can be used with different approaches such as machine learning, signal processing techniques, and deep learning to estimate the position and trajectory of devices or users in indoor environments. However, indoor localization presents several challenges, as different applications impose varying requirements in terms of accuracy, cost, scalability, and infrastructure. Consequently, no single localization approach is suitable for all indoor environments and applications. In this project, we utilize a dataset consisting of dense indoor signatures of WiFi, geomagnetic fields, and iBeacons, as well as ground truth data collected from hundreds of buildings in Chinese cities. The raw sensor data is preprocessed to reduce the effects of noise and missing observations, while the motion data is further processed to obtain a denser estimate of the user's trajectory.

The processed sensor data is visualized to examine signal patterns and their potential for localization. Beyond this visualization task, the project investigates several approaches to indoor localization. These include accelerometer-based waypoint position estimation using a Kalman filter, deep learning-based location fingerprinting using a multilayer perceptron (MLP) with iBeacon-only and multimodal sensor inputs, and sequence-based localization combining geomagnetic and IMU measurements with K-nearest neighbors (KNN) and Viterbi decoding. The objective is to evaluate the effectiveness of these approaches and determine how combining different sensor modalities and incorporating trajectory information can improve indoor localization.

Site1/B1, *site1/F2*, and *site2/F1* were selected as the main examples for each task because they contained moredata points and demonstrated greater variability across the visualisations

II. Dataset

A. Dataset Overview and Structure

The dataset from the Microsoft Indoor Location Competition 2.0 features dense indoor signatures from WiFi, geomagnetic fields, and iBeacons, along with ground truth data from hundreds of buildings in China. This project utilizes sample data available from the GitHub repository Indoor Location Competition 2.0 [1], which consists of sample data from two sites, *site1* and *site2* respectively, each including indoor traces between positions P1 and P2, recorded by a surveyor using an Android smartphone. The application on the device collects IMU-related measurements, including accelerometer, gyroscope, and rotation vector readings, as well as geomagnetic field measurements via the magnetometer, WiFi scans, and Bluetooth iBeacon results during the walk. Each floor directory also includes metadata such as a floor-plan image, floor dimensions, and GeoJSON information describing geographic structures.

Each line in the source trace file consists of a timestamp, data type, and its associated values. There are 10 different data types in the dataset representing either the ground truth or sensor values. For example, TYPE_WAYPOINT contains the x and y coordinates in meters, while TYPE_WIFI contains more detailed information such as SSID, BSSID, RSSI, frequency, and last seen timestamp. It is important to note that the values for each data type follow a specific order, which is detailed in Table I. The timestamp is Unix time in milliseconds, and for the sensor-related data types, the recorded values include fields obtained from Android's SensorEvent, such as sensor values and accuracy. The iBeacon data is obtained from Bluetooth scan records and decoded according to the iBeacon protocol.

TABLE I: Data Types and Their Values

Data Type	**Values**
TYPE_ACCELEROMETER	X, Y, Z axes; accuracy
TYPE_MAGNETIC_FIELD	X, Y, Z axes; accuracy
TYPE_GYROSCOPE	X, Y, Z axes; accuracy
TYPE_ROTATION_VECTOR	X, Y, Z axes; accuracy
TYPE_ACCELEROMETER_ UNCALIBRATED	X, Y, Z axes; uncalibrated X, Y, Z axes; accuracy
TYPE_GYROSCOPE_ UNCALIBRATED	X, Y, Z axes; uncalibrated X, Y, Z axes; accuracy
TYPE_MAGNETIC_FIELD_ UNCALIBRATED	X, Y, Z axes; uncalibrated X, Y, Z axes; accuracy
TYPE_WIFI	SSID; BSSID; RSSI; frequency; last-seen timestamp
TYPE_BEACON	UUID; Major; Minor; Tx Power; RSSI; distance; MAC address
TYPE_WAYPOINT	X and Y coordinates in meters

B. Ground Truth and Position Estimation

The ground truth values represent reference coordinates labelled by the surveyor and are used as reference points for the sensor observations. Unlike the sensor data, which is passively collected as the volunteer moves, waypoints are actively recorded. However, analysis showed that the ground truth data points are sparse, leaving gaps between consecutive waypoints. The positions within these gaps, therefore, need to be estimated from the available motion-sensor data. The accelerometer readings (TYPE_ACCELEROMETER) are used to detect steps and estimate their approximate lengths, while the rotation vector readings (TYPE_ROTATION_VECTOR) are used to estimate the direction of each step. These measurements are then used to compute the relative movement and estimate the next position. This process is repeated to estimate the trajectory between consecutive ground truth waypoints. The resulting estimated step positions provide a denser trajectory than the original waypoints, and sensor observations are then associated with the estimated step position with the closest timestamp. The details of how data is processed and augmented for geomagnetic, WiFi, and iBeacon are provided in the following section.

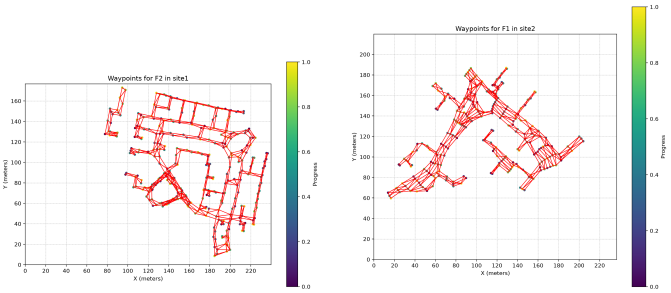


Fig. 1: Groud-truth waypoint visualization for site1, floor B1.

III. Essential Tasks

A. Visualization of Waypoints

For this task, we used the waypoint traces provided in the dataset to visualize the recorded paths across every floor in both sites. Each waypoint trace contains records with a timestamp and a corresponding ground-truth X and Y coordinates in meters. For every trace, we extracted the waypoint records in timestamp order to trace movement on the floor.



(a) Site 1, F2 (b) Site 2, F1

Fig. 2: Ground-truth waypoint visualizations for the selected site-floor combinations.

Each trace was handled independently and plotted, with consecutive waypoint coordinates connected by red lines to trace the approximate straight-line path travelled between observations. We also included a subtle color gradient on the waypoint markers to indicate the progression of the waypoint trace from beginning to end. Traces were considered separately so that the final waypoint record of one trace was not incorrectly connected to the first waypoint of another. The floor dimensions (in meters) were used to define the axis limits of each plot, and equal scaling was applied to mitigate distortion. Each figure also includes 20-meter tick intervals, grid lines, and a color bar for the progression.

The resulting figures contain overlapping paths across all waypoint traces, and some areas with a high concentration

of points within the site/floor were traversed multiple times. Some line segments between records appear long because of the sparse nature of the provided waypoint measurements. The difference in density of visualization across floors is due to the number of waypoint traces available for those floors.

B. Visualization of Geomagnetic Heatmaps

In the dataset, on each timestamp we have geomagnetic readings along with the IMU sensor readings. This gives us a way to map the geomagnetic reading to positions, since we can calculate the position where each reading was taken. In this way, we construct a stitched data mapping between magnetic field and generated waypoints. We also noticed that the ground truth waypoints do not have a corresponding sensor measurement; to fill this, we used the nearest timestamp readings. The plots show the magnitude of magnetic field strength along each axis at each waypoint as a heatmap.

We can use the heatmaps to visually analyze how much the magnetic field varies with location. This is helpful to judge how effective a fingerprinting localization technique would be. One limitation we would like to highlight is that since there are no direct mappings between field reading and spatial data, we rely on the initial generated waypoints; hence, our visualization also depends on the accuracy of the reconstructed trajectories.

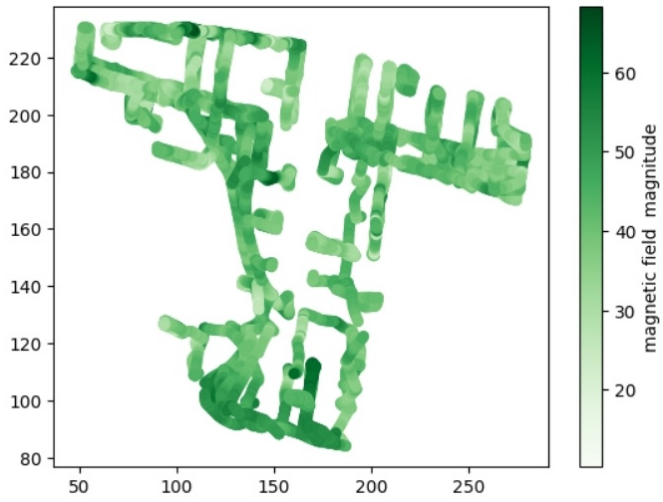
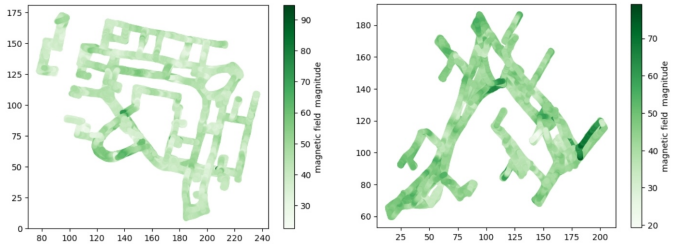


Fig. 3: Geomagnetic heatmap for site1, floor B1. Colour indicates magnetic field intensity

C. Visualization of WiFi RSS Heatmaps

As described in the earlier section, the source data is extracted from trace files, which have lines specific to a particular data type, including TYPE_WIFI, which is used to generate WiFi RSSI heatmaps. As shown in Table 1, each TYPE_WIFI record consists of information such as the timestamp of the WiFi scan result, SSID (the name of the WiFi network), BSSID (the MAC address of a WiFi access point), RSSI (Received Signal Strength Indicator expressed in dBm), the WiFi radio frequency, the last radio frequency, and the



(a) Site 1, F2 (b) Site 2, F1

Fig. 4: Geomagnetic heatmaps for the selected site-floor combinations. Colour indicates magnetic field intensity.

last seen timestamp. For the visualization, the timestamp of when the WiFi results were scanned, SSID, BSSID, and RSSI are used. It is important to retain the BSSID since it allows measurements from a specific access point to be differentiated from those of other access points.

The WiFi data offers RSSI values over time, although we lack the device's actual coordinates. To visualize the WiFi RSSI heatmaps, the WiFi BSSID, RSSI value, and its estimated position on the floor will be required. Since the ground waypoint data is relatively sparse, they don't provide a position for every WiFi scan. Accelerometer and rotation vector measurements are used to detect steps, estimate step length, identify direction, and calculate the relative movement. This information, along with the ground truth data, is used to obtain the positions of a person at each detected step between the ground truth waypoints. Each WiFi scan is mapped to an estimated step position whose timestamp is closest to the scan timestamp.

Since the visualization of a WiFi heatmap is specific to a particular access point, the data points are grouped by BSSID. When multiple observations from the same access point are associated with the same estimated position, their RSSI values are averaged and stored along with the observation count. This produces a set of RSSI measurements and their associated positions for each access point, which can be used to generate the heatmaps. Three access points are selected for the chosen site and floor, and a separate WiFi heatmap is generated for each access point. Each colored dot on the heatmap is the average signal strength at an estimated position for the selected access point. According to the color scale, the red/orange color indicates stronger received signals, and the cooler color, such as blue, indicates weaker signals.

For example, Figure 5 shows the WiFi RSSI heatmap for access point 0e:74:9c:2e:d8:36 on site 1, floor B1. The signal strength is seen to be concentrated around the central area where the red, orange, and yellow dots are visible. This suggests that the received signal strength was stronger in this region, potentially because of the presence of the access or differences in signal propagation. The weaker signals represented by blue and cyan are observed at several other locations along the surveyed paths. The signal strength can help to distinguish between locations on a map during indoor

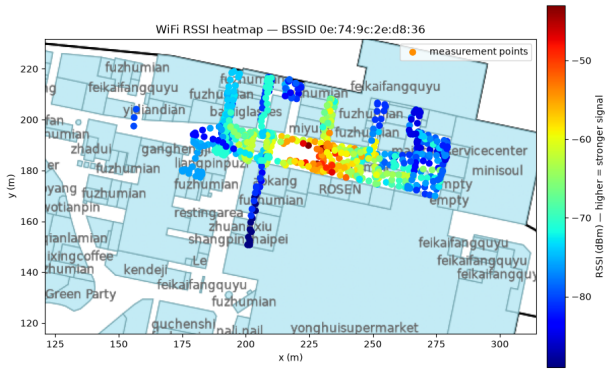
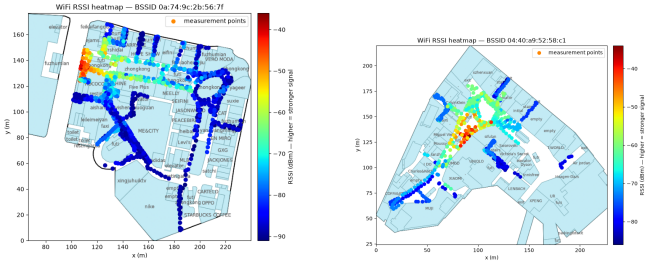


Fig. 5: WiFi RSS heatmap for site1, floor B1. Colour indicates RSSI in dBm.



(a) Site 1, F2 (b) Site 2, F1

Fig. 6: WiFi RSS heatmaps for the selected site-floor combinations. Colour indicates RSSI in dBm.

localization.

D. Visualization of iBeacon RSS Heatmaps

iBeacon devices transmit Bluetooth Low Energy (BLE) signals that get weaker with distance, making the received signal strength (RSS) a useful proxy for proximity.

For each floor, all trace files were parsed to extract TYPE_BEACON records. Each beacon reading contains the RSS value in dBm and also a unique beacon identifier formed from its UUID, Major, and Minor fields. Each reading was associated with its nearest ground-truth waypoint by matching timestamps, assigning it a physical (x, y) position on the floor plan. For each floor, we identified the most frequently observed beacon and used it as a representative signal source. Its RSS values were then plotted as a scatter plot across all recorded positions, overlaid on a coordinate frame matching the floor dimensions. For the RSS values, points with warmer colors indicate stronger signals (closer to 0 dBm), and cooler colors indicate weaker signals (approaching -100 dBm). Positions with no reading for that beacon were excluded from the plot.

Figs. 7, 8a, and 8b show the resulting heatmaps for *site1/B1*, *site1/F2*, and *site2/F1* respectively. Across all three floors, the RSS values are strongest near the beacon and decrease

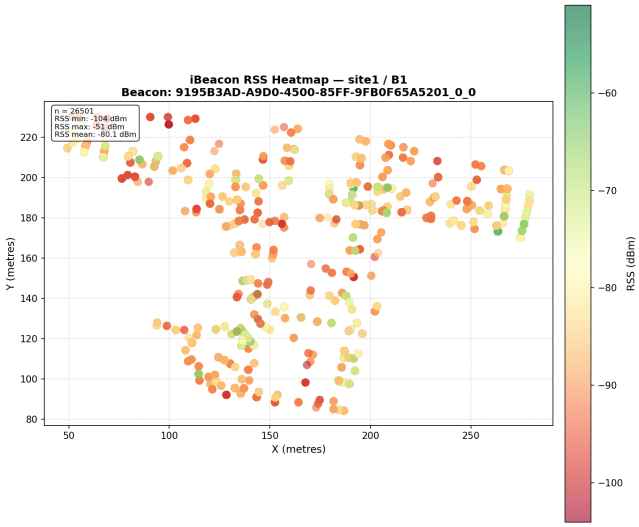
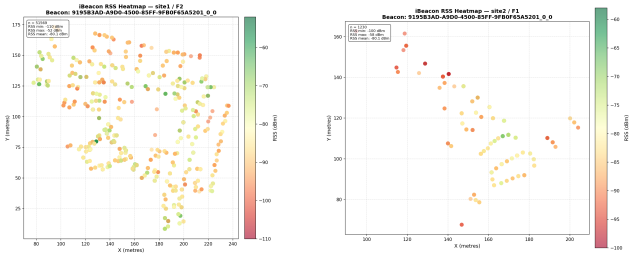


Fig. 7: iBeacon RSS heatmap for site1, floor B1. Colour indicates RSS in dBm.

with distance, which is expected behavior for BLE signals. Coverage is not uniform, and areas further from the transmitter show noticeably weaker readings. This variation in signal strength across space is what makes fingerprinting viable, as each location tends to have a distinct pattern of RSS values across multiple beacons.



(a) Site 1, F2 (b) Site 2, F1

Fig. 8: iBeacon RSS heatmaps for the selected site-floor combinations.

IV. Advanced Tasks

A. Accelerometer-Based Waypoint Position Estimation Using a Kalman Filter

A Kalman filter baseline was developed to estimate position and velocity by combining the accelerometer readings transformed to world coordinates using the rotation vector. Gravity was removed from the transformed readings, and the estimated horizontal sensor bias was subtracted before the acceleration was used for prediction. The filter tracked the X and Y positions and their corresponding velocities, with each trace starting at its first ground-truth waypoint and an initial velocity of zero. Timestamp differences between consecutive accelerometer readings were used to determine the duration of each movement update. The movement was predicted

continuously between waypoints, and every next ground-truth waypoint was used to correct accumulated position and velocity errors. At each waypoint timestamp, the predicted position was recorded and compared with the ground-truth position before the correction was applied. It resulted in only one-segment-ahead predictions because it was continually re-anchored using known waypoints. The positioning error was calculated as the Euclidean distance between the predicted and ground-truth waypoint coordinates. The mean positioning errors were 8.66 m for B1, 10.31 m for F2, and 9.86 m for F4. Small accelerometer biases also accumulated during integration, resulting in significant drift and sensitivity to the phone's orientation and sensor noise. These limitations showed that raw acceleration integration was unsuitable for reliable long-term position estimation without more frequent external position feedback.

B. Deep Learning-based Location Fingerprinting with Multimodal Fusion

1) Overview: Location fingerprinting works in two phases. In the offline phase, sensor readings are collected at known positions to build a reference map. In the online phase, a new reading is matched with this map to estimate the user's location. Instead of computing distances between fingerprints directly, we trained an MLP to predict (x, y) coordinates from a fixed-length input vector, framing it as a regression problem.

We built and compared two models, both trained on all five floors of *site1* (B1, F1, F2, F3, F4): one using only iBeacon RSS features, and another combining iBeacon, WiFi, and geomagnetic readings. Training across all floors together lets the model learn a site-wide representation rather than being limited to a single floor.

2) Feature Extraction: Each trace file was parsed to extract beacon, WiFi, and geomagnetic readings. Each sensor reading was matched to the nearest waypoint by timestamp to get its ground-truth (x, y) position. Readings at the same location were averaged to get a single RSS value per beacon or access point.

For the iBeacon-only model, the 50 most frequently observed beacons across all floors were selected. Each position was encoded as a 51-dimensional vector: one RSS value per beacon, defaulting to -110 dBm if the beacon was not seen at that location, plus a floor index (B1=0 through F4=4). The floor index was added to distinguish positions on different floors that share the same (x, y) coordinates.

For the multimodal model, the top 30 beacons, top 30 WiFi access points, and 3-axis geomagnetic readings were combined into a 64-dimensional vector (30 iBeacon + 30 WiFi + 3 geomagnetic + 1 floor index). The same -110 dBm default was used for missing RSS values. All features were standardized to zero mean and unit variance before being passed to the network.

3) Model Architecture and Training: The MLP used for both models has three fully connected layers. The input is projected to 128 units, then to 64, and the final layer outputs the (x, y) coordinate prediction directly in meters. ReLU activation and dropout (rate = 0.2) are applied after each hidden layer to prevent overfitting, and no activation is used at the output.

Before training, all features were normalized to zero mean and unit variance. We used the Adam optimizer with a learning rate of 10^{-3}, a batch size of 32, and an 80/20 train/test split, minimizing MSE loss. Prediction accuracy was then reported as the Euclidean distance in meters from the predicted position to the ground truth.

4) Results: Table II summarizes the localization accuracy of both models. The iBeacon-only model was trained for 100 epochs, while the multimodal model was evaluated at both 100 and 300 epochs.

TABLE II: Localization Accuracy: iBeacon-only vs. Multimodal (site1, All Floors)

Model	Epochs	Mean (m)	Median (m)	<5m (%)	<10m (%)
iBeacon only	100	46.45	43.66	1.3	3.6
Multimodal	100	23.32	17.82	4.8	19.1
Multimodal	300	**20.83**	**16.04**	**9.3**	**25.7**

The multimodal model at 300 epochs achieves the best performance across all metrics, with a mean error of 20.83 m and a median error of 16.04 m. This represents a 55% reduction in mean error compared to the iBeacon-only baseline (46.45 m). The proportion of predictions within 10 m also improves substantially, from 3.6% (iBeacon only) to 25.7% (multimodal, 300 epochs), suggesting that WiFi and geomagnetic readings contribute spatial information that iBeacon signals on their own do not provide.

The relatively high absolute errors across all models reflect the difficulty of multi-floor localization over a large building footprint, where RSS signals are sparse and floor-to-floor interference is significant. Nonetheless, the consistent improvement from iBeacon-only to multimodal and from 100 to 300 training epochs demonstrates the benefit of both richer feature representations and longer training.

Figs. 9 and 10 show scatter plots of predicted versus ground-truth positions for the iBeacon-only model and the multimodal model, respectively. The multimodal model produces visibly tighter clusters around the ground-truth locations, particularly in areas with dense WiFi infrastructure.

C. Sequence-based Localization using Geomagnetic and IMU Data

1) Algorithm Selection: At first, we tried using magnetic field readings to implement a direct fingerprinting technique. But the readings are not spatially unique enough to make a good identifier; even adding a sequence of readings (temporally) does not improve this. During this, we noticed that even though predicting a single location was tough, a KNN model could be used to generate a list of candidate locations. Taking the top 20 matches almost always ensures that the correct location is within the candidates. We got the initial inspiration to maintain a list of candidate trajectories from [2].From

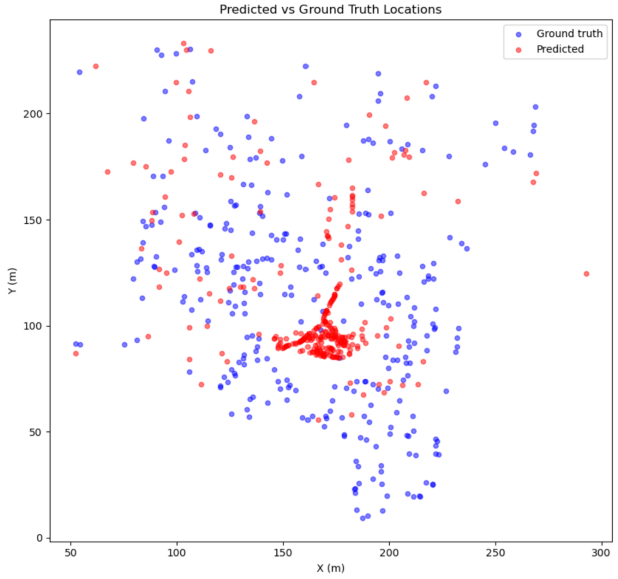


Fig. 9: Predicted vs. ground-truth positions for the iBeacon-only MLP (100 epochs).

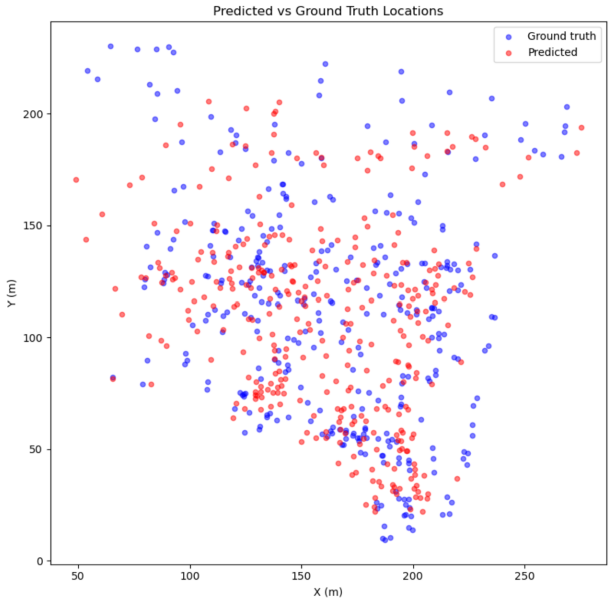


Fig. 10: Predicted vs. ground-truth positions for the multi-modal MLP (300 epochs, iBeacon + WiFi + Geomag).

here, we were inspired by dynamic programming techniques to construct a most probable trajectory, i.e., given a list of candidate locations for each point in a trajectory, using the IMU data, we can decide which combination of points is most likely to be correct. Since consecutive positions are temporally correlated, a spatial transition cost is introduced to enforce trajectory consistency. The resulting sequence optimization problem is solved using the Viterbi algorithm, which identifies the minimum-cost path through the candidate locations.

2) Sequence Construction and Preprocessing: Instead of using individual sensor measurements, we considered a sequence of 20 consecutive observations. Each observation contains three geomagnetic measurements and nine IMU measurements, resulting in 60-dimensional magnetic and 180-dimensional IMU feature vectors, respectively. The magnetic and IMU features were standardized separately using statistics computed from the training data.

3) Candidate Generation: For each test sequence, $K = 20$ candidate locations were retrieved using the K-nearest-neighbour search in the standardized magnetic feature space. The magnetic distance between a test sequence t and training sequence i is defined as

$$d_{\mathrm{mag}}(t,i) = \|\mathbf{x}_t^{\mathrm{mag}} - \mathbf{x}_i^{\mathrm{mag}}\|_2 . \tag{1}$$

4) Observation Cost: The IMU measurements are incorporated into the candidate evaluation through a corresponding Euclidean distance,

$$d_{\mathrm{imu}}(t,i) = \left\|\mathbf{x}_t^{\mathrm{imu}} - \mathbf{x}_i^{\mathrm{imu}}\right\|_2 . \tag{2}$$

The observation cost for candidate i at time t is then defined as

$$E_t(i) = d_{\mathrm{mag}}(t,i) + \beta d_{\mathrm{imu}}(t,i), \tag{3}$$

where $\beta = 0.25$ controls the contribution of the IMU measurements. Thus, candidate selection considers both the magnetic fingerprint and the associated inertial measurements rather than relying exclusively on magnetic similarity.

5) Temporal Inference using Viterbi Decoding: Independent selection of the minimum-cost candidate at each timestamp can result in physically inconsistent trajectories. To incorporate trajectory continuity, a transition cost is defined between consecutive candidate locations:

$$T(j,i) = \gamma \|\mathbf{p}_{t-1,j} - \mathbf{p}_{t,i}\|_2 , \tag{4}$$

where $\mathbf{p}_{t,i}$ denotes the physical coordinates of candidate i at time t, and $\gamma = 0.0225$ controls the transition penalty.

The cumulative cost of reaching candidate i at time t is computed using the Viterbi recurrence

$$V_t(i) = E_t(i) + \min_j \left[V_{t-1}(j) + T(j,i)\right]. \tag{5}$$

The minimum-cost path is subsequently recovered by backtracking from the final timestep. Samples are processed in timestamp order within each trajectory, ensuring that the temporal model operates on the actual measurement sequence.

6) Evaluation and Discussion: The proposed method achieved a mean localization error of approximately $\mathbf{1.49}\,\mathrm{m}$ and a median error of $\mathbf{0.59}\,\mathrm{m}$. The 90th and 95th percentile errors were approximately $1.00\,\mathrm{m}$ and $1.40\,\mathrm{m}$, respectively. Overall, 89.92% of predictions were within 1 m, 96.69% within 5 m, and 97.59% within 10 m.

Despite the strong overall performance, a small number of large-error predictions remained. Out of 665 test samples,

only 16 (2.41%) had errors exceeding 10 m. However, these failures were substantial, with a mean error of approximately 34.0 m and a maximum error of 84.55 m. The large difference between the 95th-percentile error and the maximum error indicates that the mean error is strongly affected by a small number of catastrophic localization failures. These errors mostly corresponded to the ranking algorithm choosing a wrong candidate point (even when the correct point is among the candidates), due to the overall similarity between the sensor readings (both IMU and magnetic) between these locations. We tried incorporating other sensors such as Wi-Fi/beacons to reduce these specific errors, but the overall performance degraded as the algorithm cannot confidently identify points of failure.

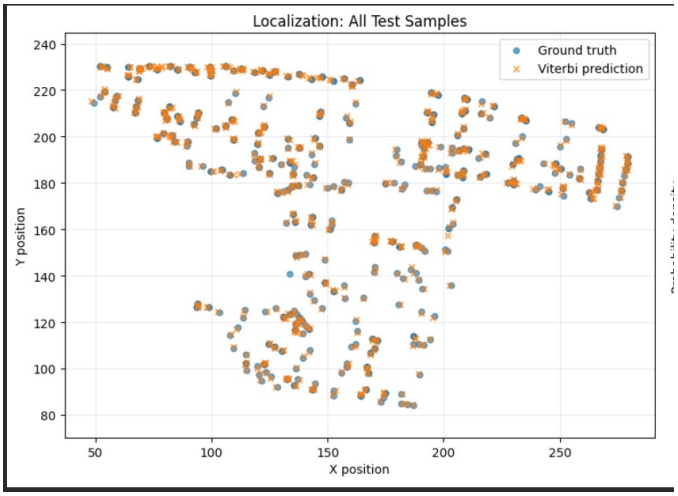


Fig. 11: Predicted vs. ground-truth positions for viterbi algorithm.

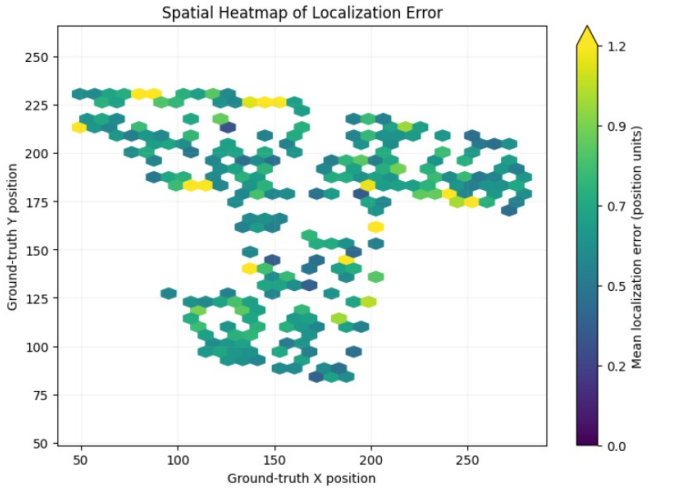


Fig. 12: Spatial error distribution for viterbi algorithm(including 95 percentile only)

V. Conclusion

This project evaluated multiple approaches to smartphone-based indoor localization using the Indoor Location Competition 2.0 dataset. The essential tasks involved visualizing waypoint traces and signal heatmaps for geomagnetic, WiFi, and iBeacon data, which helped establish an understanding of how each signal varies across indoor spaces.

For the advanced tasks, the Kalman filter served as a baseline but suffered from drift over longer trajectories. The MLP fingerprinting model improved with multimodal inputs, reducing mean error from 46.45 m to 20.83 m when WiFi and geomagnetic features were added. However, the best results came from the sequence-based Viterbi approach, which achieved a mean error of 1.49 m and a median of 0.59 m, showing that using a sequence of sensor readings over time to track movement gives significantly better results than predicting location from a single set of measurements.

Group Member Contributions

All four group members contributed equally to this project. Each member was involved across all portions of the work, including data preprocessing, visualization, model implementation, and report writing.

References

[1] Indoor Location Competition 2.0 - https://github.com/location-competition/indoor-location-competition-20

[2] T. Van de Wiele, D. Gordeev, and A. Haartveit, "1st Place Solution – Track me if you can," Kaggle, Indoor Location & Navigation competition, May 25, 2021.[Online]. Available: https://www.kaggle.com/competitions/indoor-location-navigation/writeups/track-me-if-you-can-1st-place-solution-track-me-if

Appendix

The source code developed for this project is available in the following GitHub repository:
github.com/comicsans-02/indoor_localization_ai6128

The repository contains the implementations of the indoor localization techniques and the code used to generate the visualizations presented.